Reg. No. | | | | | | | | | | |

[18ME64]

SIXTH SEMESTER B.TECH EXAMINATION, JULY 2022

MECHANICAL VIBRATIONS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

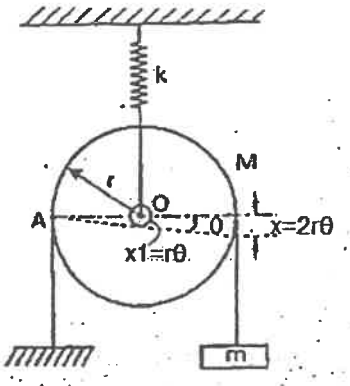
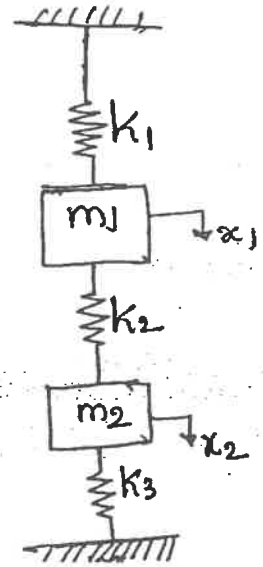
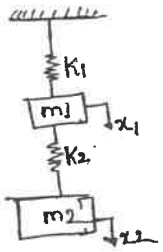
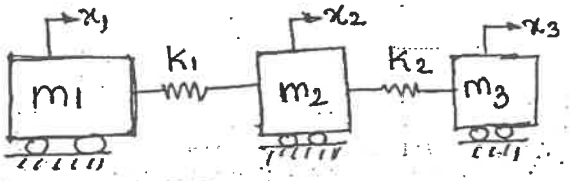
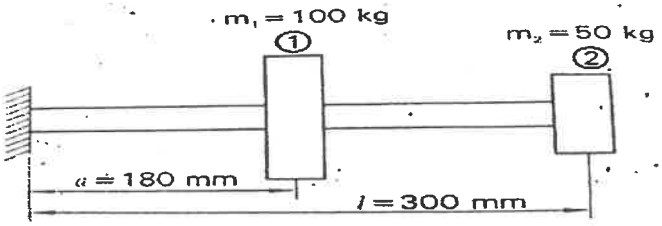
1.	**a.**	Determine the equation of motion and natural frequency of the system by newton's method shown in Fig. 1a.  Fig. 1a	**08**
	b.	Explain different types of damping.	**06**
2.	**a.**	Define transmissibility and derive an expression for transmissibility ratio and phase angle.	**10**
	b.	Explain force and motion transmissibility.	**04**
3.		Find the natural frequency and mode shapes of the system shown in figure.3 for $K_1=K$, $K_2=K$, $K_3=K$ and $m_1=m$, $m_2=m$. 	**14**

		Fig.3	
		OR	
4.		Find the natural frequency and mode shapes of the system shown in figure.4 for $K_1=2k$, $k_2=k$ and $m_1=2m$, $m_2=m$.  Fig.4	**14**
5.		Determine the natural frequencies of the system shown in the fig. 5 by Holzer's method. Take $m_1=2Kg$, $m_2=4kg$, $m_3=4kg$ $K_1=5N/m$, $k_2=10N/m$  Fig.5	**14**
		OR	
6.	**a.**	Find the lowest natural frequency of the vibration for the system shown in fig.6a by Rayleigh's method $E:1.96X10^{11}$ N/m^2 ; $I=4X10^{-7}$ m^4  Fig. 6a	**07**
	b.	Explain the terms Multi-DOF system, Influence co-efficient and Flexibility matrix.	**07**
7.	**a.**	Explain any one frequency measuring instrument with neat sketch.	**07**
	b.	A seismic instrument with a natural frequency of 5Hz is used to measure the vibration of a machine operating at 150rpm. The relative displacement of the seismic mass as read from the instrument is 0.05mm, neglecting air damping, determine the amplitude of vibration of the machine.	**07**
		OR	
8.	**a.**	Determine the limiting frequency ω for an accelerometer with 2.5% error having damping factor ϵ =0.7 and natural frequency 80 Hz.	**07**
	b.	With neat sketches explain in detail any two condition monitoring techniques used in the vibrational analysis.	**07**